

# AI Riviera PostgreSQL Guide

How the tables store municipal documents, political objects, people and financial data

**A relational database is a collection of tables connected by identifiers. The important idea is simple: store each fact once, then link related rows instead of copying the same data everywhere.**

|             |                                                           |
|-------------|-----------------------------------------------------------|
| TABLE       | A named collection of similar records                     |
| ROW         | One record, such as one document or one person            |
| COLUMN      | One attribute, such as title, date or status              |
| PRIMARY KEY | The unique identifier of a row                            |
| FOREIGN KEY | A stored identifier that points to a row in another table |

# 1. How table relationships work

Imagine one municipal PDF. The documents table stores the PDF once. The PDF is divided into several searchable passages, so the document\_chunks table stores many rows that all point back to that document.



## Primary key (PK)

A unique value that identifies one row. `documents.id` identifies one document.

## Foreign key (FK)

A value copied into another table to create a link. `document_chunks.document_id` points to `documents.id`.

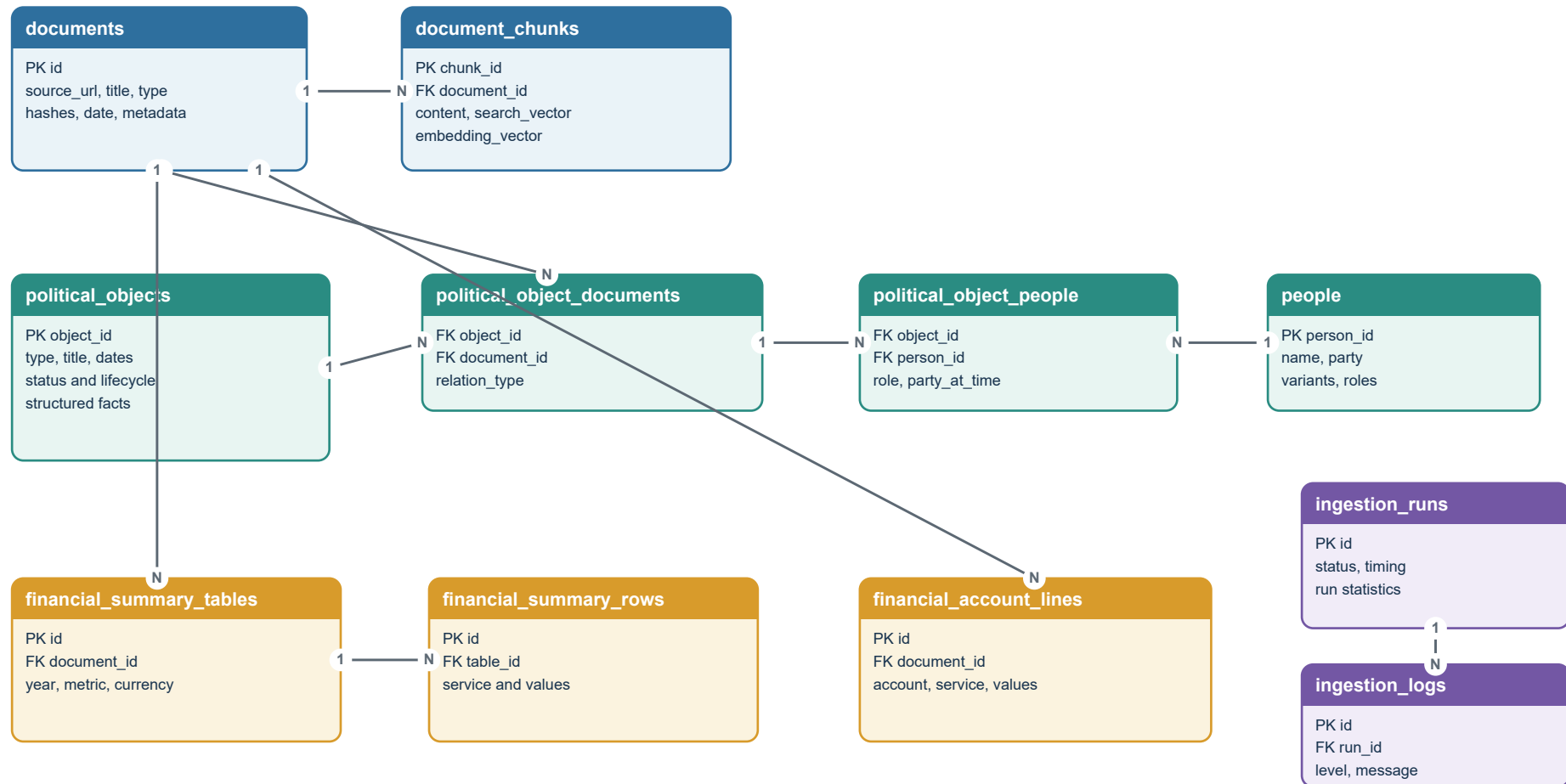
## One-to-many (1:N)

One document can contain many chunks, but each chunk belongs to one document.

**Deletion rule: the schema uses ON DELETE CASCADE for chunks. If a document is deleted, its chunks are automatically deleted too, preventing orphaned data.**

## 2. AI Riviera database map

The schema has four areas: document retrieval (blue), political knowledge (green), financial extraction (gold), and ingestion monitoring (purple). Labels 1 and N show one-to-many relationships.



The city value is currently stored directly in several tables. There is no separate cities table in the active schema.

### 3. Core document and search tables

| Table           | Purpose                                           | Important fields                                                | Relationship                                          |
|-----------------|---------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------|
| documents       | One row per official source document.             | id, city, source_url, title, doc_type, hashes, date, metadata   | Parent of chunks, political links and financial rows. |
| document_chunks | Searchable passages extracted from each document. | chunk_id, document_id, content, search_vector, embedding_vector | Many chunks belong to one document.                   |

#### How a question uses these tables

- The user asks a question.
- PostgreSQL searches document\_chunks.search\_vector for French keyword matches.
- When real embeddings are added, pgvector searches embedding\_vector for passages with similar meaning.
- The winning chunks point through document\_id to the parent document.
- The answer shows the document title and official source URL so the user can verify it.

**Current detail: embedding is also stored as JSONB, while embedding\_vector is the optional pgvector column used for efficient cosine-similarity search. Real BGE-M3 embeddings are a proposed improvement; the current fallback vectors are not strong semantic embeddings.**

#### Why hashes are stored

document\_hash and content\_hash help detect changes. If a PDF has not changed, the ingestion pipeline can avoid processing and indexing it again.

## 4. Political objects and many-to-many relationships

A motion can have several authors, and one person can author several motions. That is a many-to-many relationship. Relational databases represent it with a junction table.

| Table                      | What one row represents                                          | Key relationship                                        |
|----------------------------|------------------------------------------------------------------|---------------------------------------------------------|
| political_objects          | One motion, postulate, interpellation or other political object. | Referenced by the two junction tables.                  |
| people                     | One normalized person, with name variants and party information. | Connected to objects through political_object_people.   |
| political_object_people    | One person's role on one political object.                       | object_id → political_objects; person_id → people.      |
| political_object_documents | One document's relationship to one political object.             | object_id → political_objects; document_id → documents. |

### Example

|                         |               |                            |
|-------------------------|---------------|----------------------------|
| political_objects       | OBJ-17        | Motion: Safer school route |
| people                  | P-04          | Alice Martin               |
| people                  | P-09          | Marc Dupont                |
| political_object_people | OBJ-17 + P-04 | role = author              |
| political_object_people | OBJ-17 + P-09 | role = co-author           |

The names are not copied into every political object. The junction rows store the relationships, which reduces duplication and makes updates safer.

## 5. Financial and ingestion tables

| Table                    | Purpose                                                                   | Relationship                                    |
|--------------------------|---------------------------------------------------------------------------|-------------------------------------------------|
| financial_summary_tables | Represents one extracted summary table from a budget or account document. | Many summary tables can belong to one document. |
| financial_summary_rows   | Stores the individual services and values inside a summary table.         | Many rows belong to one summary table.          |
| financial_account_lines  | Stores detailed account lines for structured financial questions.         | Many account lines can belong to one document.  |
| ingestion_runs           | Records one complete import or scheduled update.                          | Parent of ingestion logs.                       |
| ingestion_logs           | Stores warnings, errors and information generated during a run.           | Many logs belong to one run.                    |

### How to read the schema quickly

- Start with the table's primary key: it tells you what makes a row unique.
- Look for columns ending in `_id`: they are usually foreign keys pointing to another table.
- A foreign key is normally on the many side of a one-to-many relationship.
- A table containing two foreign keys often acts as a junction table for a many-to-many relationship.
- JSONB holds flexible metadata; normal columns hold facts that need reliable filtering and sorting.
- Indexes do not create relationships. They make searches and joins faster.

**Mental model: documents are the evidence, chunks make that evidence searchable, political tables organize exact facts, financial tables make numbers queryable, and ingestion tables tell us whether the data pipeline worked.**